BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 129 · Lesson 129 · Dictionary Compression and Repeated Substrings

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. RLE is effective on what kind of input?

Q2. How many bits are in three bytes?

Q3. Which memory is typically volatile: RAM or ROM?

Q4. What is the binary result of 1 + 1 in a full adder with carry-in 0?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

dictionary coding — Encodes recurring sequences by referring to dictionary entries.

token — Short code representing a longer sequence.

overlap — Repeated sequence used by more than one part of the source.

lossless — Reconstruction is exact when dictionary and token mapping are known.

New learning

Find recurring text — Locate useful repeated words or substrings.

Build dictionary — Assign tokens to recurring sequences and store mapping.

Encode and decode — Replace sequences with tokens; decode using the same mapping.

Worked example

Use dictionary {#1:'COMPUTER', #2:'SCIENCE'} to encode 'COMPUTER SCIENCE COMPUTER'.

1. Two dictionary entries — Predict the first step from the prompt.

2. #1 #2 #1 — Explain the intermediate result or decision.

3. Restore original phrase — Treat spaces as retained separators: #1 #2 #1. Decoder substitutes entries to recover 'COMPUTER SCIENCE COMPUTER'. Include dictionary storage when calculating total size.

We Do · Practise with guidance

Dictionary {A:'ABCD'}; encode 'ABCDABCDXYZABCD' and discuss the result.

ABCD repeats three times

Predict the first step from the prompt.

Token A for ABCD

Explain the intermediate result or decision.

Count dictionary overhead

AAXYZA, using single-character token A and leaving XYZ as literals under this toy scheme. Original 15 characters become 6 token/literal characters, plus dictionary overhead; real formats require unambiguous token conventions.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Explain dictionary compression and why the decoder needs the dictionary. [6]

Question 2

Using entries {1:'RED',2:'BLUE'}, encode 'RED BLUE RED RED' and decode '2 1'. [6]

Question 3

Evaluate dictionary coding for a document with frequent terms versus already-compressed random-looking data. [8]

Exit Ticket · Answer before checking

Dictionary coding requires permanently discarding repeated words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Long consecutive runs of identical symbols.

Q2. 24.

Q3. RAM.

Q4. Sum 0, carry-out 1.

You Do pages · Independent questions

Q1. Repeated substrings replaced by shorter references [3]; mapping resolves tokens for exact restoration [3].

Q2. 1 2 1 1 [3]; BLUE RED [3], with token separators assumed.

Q3. Repeated sequences help [2], tokens shorten recurring data [2], dictionary overhead matters [2], little redundancy in high-entropy data and justified choice [2].

Exit Ticket · Model answer

False. The dictionary allows exact replacement when decoding.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: